REVIEW · REVIEW

Week 1 recall day

Answers and worked solutions · Day 09 · Sun 1 Nov · pack R1

Part A — Machines 2024 mid-sem test (timed, 60 min)

Q1 Fill-in-the-Blank Questions [4]

(a) [2] The US mains is 120 V, 60 Hz. The quoted value is rms.

- Peak: $\hat{V} = \sqrt{2} \times 120 = \mathbf{169.7\ V}$
- rms: $\mathbf{120\ V}$
- Period: $T = 1/60 = \mathbf{16.67\ ms}$
- 90° is a quarter period: $T/4 = \mathbf{4.167\ ms}$

(b) [2] The star load is on a 400 V supply. A supply voltage is a line voltage unless stated.

- Line voltage $V_L = \mathbf{400\ V}$
- Phase voltage $V_1 = 400/\sqrt{3} = \mathbf{230.9\ V}$
- Line current $I_L = \mathbf{15\ A}$
- Phase current $I_1 = I_L = \mathbf{15\ A}$ (star, so line current equals phase current)

OFFICIAL: 169.7 V, 120 V, 16.67 ms, 4.167 ms; 400 V, 230.9 V, 15 A, 15 A. **Agrees.**

MARKING: ½ mark per blank. You lose the mark if you write “120 V peak” or if you divide the current by $\sqrt{3}$ in star.

Q2 Real and Reactive Power [4]

GIVEN: $P = 1200$ W, $Q = +400$ VAr (absorbing, so inductive).

(a) [2]

$$S = P + jQ = \mathbf{1200 + j400\ VA} = \mathbf{1265\angle +18.43^\circ\ VA}$$

The apparent power is $|S| = \sqrt{1200^2 + 400^2} = \mathbf{1265\ VA}$.

(b) [2] For the power triangle, draw $P = 1200$ W horizontally to the right and $Q = 400$ VAr vertically **up** from its tip (positive Q). S is the hypotenuse at $+18.43^\circ$.

$$\text{pf} = \frac{P}{S} = \frac{1200}{1265} = \mathbf{0.9487\ lag}$$

It lags because $Q > 0$: the load absorbs reactive power, as an inductor does. The current angle is $\phi = -18.43^\circ$.

OFFICIAL: $1200 + j400$ VA, $1265\angle +18.44^\circ$ VA, 1265 VA, 0.9487 lag. **Agrees** (18.43° vs 18.44° is rounding).

MARKING: you lose ½ mark if the pf has no “lag”.

Q3 AC Circuit Analysis, parallel RL [8]

(a) [4] In a parallel circuit, the voltage is common, so it is the reference.

1. $V_1 = \mathbf{240\angle 0^\circ\ V}$
2. $I_R = V_1/R = 240/120 = \mathbf{2\angle 0^\circ\ A}$
3. $I_L = V_1/(jX_L) = 240\angle 0^\circ/60\angle 90^\circ = \mathbf{4\angle -90^\circ\ A}$
4. $I_1 = I_R + I_L = 2 - j4 = \mathbf{4.472\angle -63.43^\circ\ A}$ (KCL with phasors, not magnitudes)

(b) [2] The template draws V_1 pointing straight up, so use that as the reference direction. Draw I_R along V_1 with length 2. Draw I_L at 90° clockwise from V_1 (lagging, so pointing right on the template) with length 4. Draw I_1 as the diagonal of the rectangle, 63.43° clockwise from V_1. Label all three.

(c) [2]

$$S = V_1 I_1^* = 240 \times (2 + j4) = 480 + j960\ \text{VA}$$

So $P = \mathbf{480\ W}$ and $Q = \mathbf{960\ VAr}$.

Check: $V^2/R = 240^2/120 = 480$ W and $V^2/X_L = 240^2/60 = 960$ VAr. Use V^2/R here, not $I_1^2 R$, because I_1 does not flow through R.

OFFICIAL: $240\angle 0^\circ$, $2\angle 0^\circ$, $4\angle -90^\circ$, $4.472\angle -63.43^\circ$ A; 480 W, 960 VAr. **Agrees.**

Q4 Electromagnetics [7]

GIVEN: $A = 5\ \text{cm}^2 = 5 \times 10^{-4}\ \text{m}^2$, $\mu_R = 4000$, total iron path $l_C = 0.25$ m, $N = 5$, $I = 3.5$ A (so $NI = 17.5$ At), and each core has $m = 0.3$ kg.

(a) No gap [2]. All of the mmf is across the iron. $NI = H_C l_C$ and $B = \mu_R \mu_0 H_C$, so

$$B = \frac{\mu_R \mu_0 NI}{l_C} = \frac{4000 \times 4\pi \times 10^{-7} \times 17.5}{0.25} = \mathbf{0.3519\ T}$$

(b) Paper 0.07 mm thick [3]. The flux crosses the paper twice on each loop (once at each joint), so the total gap is $l_G = 2 \times 0.07 = 0.14$ mm.

$$NI = B\left(\frac{l_C}{\mu_R \mu_0} + \frac{l_G}{\mu_0}\right) = B\,(49.74 + 111.4) \times 10^3$$

$$B = \frac{17.5}{161.1 \times 10^3} = \mathbf{0.1086\ T}$$

The paper has more than twice the reluctance of the whole iron path.

(c) Minimum B to hold the lower core [2]. Weight $= mg = 0.3 \times 9.81 = 2.943$ N. Both pole faces pull, so the force area is $2A = 10\ \text{cm}^2 = 10^{-3}\ \text{m}^2$.

$$F = \frac{B^2}{2\mu_0}(2A) \ \Rightarrow\ B = \sqrt{\frac{2\mu_0 mg}{2A}} = \sqrt{\frac{2 \times 4\pi \times 10^{-7} \times 2.943}{10^{-3}}} = \mathbf{0.0860\ T}$$

`OFFICIAL:` 0.3519 T, 0.1086 T, 0.08600 T. **Agrees.**

`MARKING:` using $l_G = 0.07$ mm gives $B = 0.1845$ T and loses about 1 mark. Using area A instead of $2A$ in (c) gives 0.1216 T and loses 1 mark.

Q5 Magnetic Circuits [5]

`GIVEN:` 240 V/48 V, 50 Hz, $A = 40\ \text{cm}^2 = 4 \times 10^{-3}\ \text{m}^2$, $\hat{B}_{\max} = 1.6$ T. The path length and μ_R are not needed in this question.

(a) [2] Back-emf equation, $V = 4.44 f N \hat{B} A$:

$$N_1 = \frac{V_1}{4.44 f \hat{B} A} = \frac{240}{4.44 \times 50 \times 1.6 \times 4 \times 10^{-3}} = 168.9 \ \Rightarrow\ \mathbf{169\ turns}$$

Round **up**, because $\hat{B}$ is a ceiling. With 169 turns, $\hat{B} = 1.599$ T.

(b) [2] Wire area $A_w = \pi(0.25 \times 10^{-3})^2 = 1.963 \times 10^{-7}\ \text{m}^2$ (use the radius, 0.25 mm).

$$R = \frac{l}{\sigma A_w} = \frac{40}{59 \times 10^6 \times 1.963 \times 10^{-7}} = \mathbf{3.453\ \Omega}$$

(c) [1] $N_2 = N_1 V_2 / V_1 = 169 \times 48/240 = 33.8 \ \Rightarrow\ \mathbf{34\ turns}$

`OFFICIAL:` 169 turns, 3.453 Ω, 34 turns. **Agrees.**

Q6 Transformer Parameters [4]

`GIVEN:` a star-connected three-phase transformer. SC test: 710 V, 390 A, 140 kW. These are line voltage, line current and total power. The OC data is not needed for R and X.

Method (M2 Playbook D, series model). Convert to per-phase quantities first, then use the series RL formulas. In the SC test the shunt branch is negligible, so the meter sees the series $R + jX$.

1. $V_1 = 710/\sqrt{3} = 409.9$ V, $I_1 = 390$ A, $P_1 = 140/3 = 46.67$ kW
2. $S_1 = V_1 I_1 = 159.9$ kVA, $Q_1 = \sqrt{S_1^2 - P_1^2} = 152.9$ kVAr
3. $R = P_1/I_1^2 = 46\,667/390^2 = \mathbf{0.3068\ \Omega}$
4. $X = Q_1/I_1^2 = 152\,906/390^2 = \mathbf{1.005\ \Omega}$

Check: $|Z| = V_1/I_1 = 1.051\ \Omega$, and $\sqrt{1.051^2 - 0.3068^2} = 1.005\ \Omega$.

OFFICIAL: $X = 1.005\ \Omega$, $R = 0.3086\ \Omega$, with $V_1 = 409.9$ V, $I_1 = 390$ A, $P_1 = 46.67$ kW and $Q_1 = 152.9$ kVAr.

Disagreement with the official sheet. The official intermediate values are the same as these, but $46\,670/390^2 = 0.3068\ \Omega$, not $0.3086\ \Omega$. The official sheet has transposed two digits. X agrees. If you wrote 0.3068 Ω, you are right.

Optional (not asked). The OC data gives the shunt branch: $V_1 = 3002$ V, $P_1 = 16.33$ kW, $R_C = V_1^2/P_1 = 551.8\ \Omega$, $Q_1 = 134.1$ kVAr and $X_M = V_1^2/Q_1 = 67.21\ \Omega$ (parallel model, so use V^2/P and V^2/Q).

MARKING: forgetting the $\sqrt{3}$ and the $/3$ gives $R = 0.920\ \Omega$ and loses about 2 marks.

Q7 Transformer Analysis [4]

GIVEN: three-phase, 14 kV : 135 kV, 250 MVA. The load has $|S| = 150$ MVA and **generates** 35 MVAr, so $Q = -35$ MVAr (capacitive). The output is at rated voltage. Refer to the primary (14 kV side), assuming star.

1. **Referred phase voltage.** At rated output, $V_2' = 14\ \text{kV}/\sqrt{3} = \mathbf{8.083\angle 0^\circ\ kV}$.
2. **Power triangle.** $P = \sqrt{150^2 - 35^2} = 145.9$ MW.
   $S = 145.9 - j35 = 150\angle -13.49^\circ$ MVA. Equivalently, $\sin(-\phi) = Q/S = -35/150$, so $\phi = +13.49^\circ$.
3. **Current magnitude.**
   $I_2' = S/(3V_2') = S/(\sqrt{3}V_L) = 150 \times 10^6/(\sqrt{3} \times 14 \times 10^3) = 6.186$ kA.
4. **Current angle.** $S = 3V_2'I_2'^*$, so $\angle I_2' = -\angle S = +13.49^\circ$. The current leads, as it should for a load that generates VAr.

$$I_2' = \mathbf{6.186\angle +13.49^\circ\ kA}$$

OFFICIAL: $V_2' = 8.083\angle 0^\circ$ kV, $I_2' = 6.186\angle +13.49^\circ$ kA. **Agrees.**

MARKING: a negative angle (treating the load as lagging) loses about 1 mark. Using 250 MVA (the rating) instead of 150 MVA (the load) gives 10.31 kA and loses about 2 marks.

Q8 Short-Answer Question [4]

The question says “words only”, so no equations or sketches.

Ideal transformer [2].

- The output voltage does not change. It is set only by the input voltage and the turns ratio, and the load current does not affect it.

- The output current halves. With the voltage fixed, doubling the load resistance halves the current (Ohm's law). The input current also halves, because it is the output current scaled by the turns ratio.

Real transformer [2].

- The output voltage rises slightly. A real transformer has series winding resistance and leakage reactance. When the load current falls, the voltage drop across these series elements falls, so more of the supply voltage reaches the output (the regulation improves).
- The output current still falls, but to slightly more than half the original value, because the output voltage is now slightly higher.

`OFFICIAL POINTS:` the same four points. Each is worth 1 mark. You only get the mark for each "what" if you also give the "why".

Part B — Mixed recall (45 min)

B1.

$\omega_0 = 2\pi \times 700/1000 = 1.4\pi$. Subtract 2π: $\omega_0 = -0.6\pi$. Apparent frequency $F_a = 700 - 1000 = -300$ Hz, so the output is $\cos(2\pi \cdot 300t)$, a **300 Hz** tone. The sign is irrelevant for a cosine. It would flip a sine.

B2.

$S = 230 \times 5 = 1150$ VA. $P = S\cos\phi = \mathbf{920\ W}$.
$Q = S\sin(-\phi) = 1150 \times 0.6 = \mathbf{+690\ VAr}$, positive because it lags (inductive).

B3.

$p \approx N_A = 10^{16}\ \mathrm{cm}^{-3}$. $n = n_i^2/p = 2.25 \times 10^{20}/10^{16} = \mathbf{2.25 \times 10^4\ cm^{-3}}$ (mass-action law, $n_0 p_0 = n_i^2$).

B4.

- **OC test** gives the shunt branch, R_C and X_M. Use the **parallel** model: $R_C = V^2/P$, $X_M = V^2/Q$. The current is small, so the series drop is negligible.
- **SC test** gives the series branch, R and X. Use the **series** model: $R = P/I^2$, $X = Q/I^2$. The voltage is small, so the shunt branch draws almost nothing.
- In three-phase, convert to per-phase quantities first.

B5.

$V_{o,peak} = 12\sqrt{2} - 0.7 = \mathbf{16.27\ V}$. Without a capacitor, the PIV is the peak secondary voltage, $12\sqrt{2} = 16.97$ V. (With a smoothing capacitor, the PIV rises to about $2V_{peak}$.)

B6.

- **Linear:** yes (no offset, no nonlinear function of x).
- **Time-invariant:** no. The explicit n multiplies the input. Delaying the input gives $n\,x(n-d)$, but $y(n-d) = (n-d)\,x(n-d)$.
- **Causal:** yes (only $x(n)$).
- **Stable:** no. Take $x(n) = u(n)$, which is bounded. Then $y(n) = n\,u(n)$, which is unbounded.

B7.

$Q_{old} = P\tan(\cos^{-1} 0.7) = 2000 \times 1.0202 = 2040$ VAr. For unity pf, $Q_C = -2040$ VAr.

$$C = \frac{|Q_C|}{\omega V^2} = \frac{2040}{2\pi \times 50 \times 240^2} = \mathbf{112.8\ \mu F}$$

The supply current falls from 11.90 A to 8.333 A. P does not change.

B8.

$\Delta V_D = nV_T \ln(I_2/I_1) = 25.9\ \text{mV} \times \ln 4 = \mathbf{35.9\ mV}$. I_S cancels.

B9.

$y(n) = \{\underline{1}, 3, 3, 2\}$ for $n = 0, \ldots, 3$. Length $2 + 3 - 1 = 4$. It starts at $0 + 0 = 0$. Check: the sum of y is 9, which equals $(\sum x)(\sum h) = 3 \times 3$.

B10.

$P = \sqrt{3}V_L I_L \cos\phi = \sqrt{3} \times 415 \times 10 \times 0.85 = \mathbf{6.110\ kW}$.
$Q = \sqrt{3}V_L I_L \sin(-\phi) = \mathbf{+3.787\ kVAr}$. Using $3V_L I_L$ would overstate both by $\sqrt{3}$.

B11.

$I_B = (12 - 0.7)/470\text{k} = 24.04\ \mu$A. $I_C = \beta I_B = 2.404$ mA.
$V_{CE} = 12 - 2.404\text{m} \times 2.2\text{k} = \mathbf{6.711\ V}$. Since $V_{CE} > V_{CE,sat}$ (0.2 V), it is in **forward-**

active mode, so the assumption holds.

B12.

$X(\omega) = \dfrac{1}{1 - ae^{-j\omega}}$, which exists for $|a| < 1$ (absolutely summable). At $\omega = 0$, $|X| = 1/(1 - 0.5) = \mathbf{2}$. At $\omega = \pi$, $e^{-j\pi} = -1$, so $|X| = 1/1.5 = \mathbf{0.6667}$. It is lowpass.

B13.

$$\mathcal{R} = \frac{l}{\mu_R \mu_0 A} = \frac{0.3}{2000 \times 4\pi \times 10^{-7} \times 4 \times 10^{-4}} = \mathbf{2.984 \times 10^5\ At/Wb}.$$

$L = N^2/\mathcal{R} = 200^2/2.984 \times 10^5 = \mathbf{0.1340\ H}$.

B14.

> **CHECK:** remove the Zener and find the open-circuit load voltage, $15 \times 1000/1200 = 12.5$ V. That is greater than $V_Z = 6.2$ V, so the Zener is in breakdown and regulating.

Currents: $I_R = (15 - 6.2)/200 = 44.0$ mA. $I_L = 6.2/1000 = 6.2$ mA.
$I_Z = I_R - I_L = \mathbf{37.8\ mA}$.

With a 100 Ω load, the open-circuit voltage would be 5 V, which is less than 6.2 V. The Zener would be off and $V_o = 5$ V.

B15.

Use the causal exponential result: $a^n u(n) * b^n u(n) = \dfrac{a^{n+1} - b^{n+1}}{a - b} u(n)$.

$$y(n) = \frac{(\frac{1}{2})^{n+1} - (\frac{1}{3})^{n+1}}{\frac{1}{6}} u(n) = \left[3\left(\tfrac{1}{2}\right)^n - 2\left(\tfrac{1}{3}\right)^n\right] u(n)$$

Check: $y(0) = 1$ and $y(1) = 3/2 - 2/3 = 0.8333$. Direct convolution gives the same values.

B16.

$a = 2400/240 = 10$. $Z' = a^2 Z = 100 \times 2 = \mathbf{200\ \Omega}$. Voltages scale by a, currents by $1/a$, and impedances by a^2.

B17.

$I_B = (5 - 0.7)/20\text{k} = 0.215$ mA. Assuming active mode gives $\beta I_B = 21.5$ mA. But $I_{C,sat} = (10 - 0.2)/1\text{k} = 9.8$ mA, which is smaller. So the active assumption fails (V_{CE} would be negative). The BJT is **saturated**, with $I_C = \mathbf{9.8\ mA}$ and $V_{CE} = 0.2$ V. Equivalently, $I_B > I_{C,sat}/\beta = 98\ \mu$A.

B18.

$H(\omega) = 0.5 + 0.5e^{-j\omega} = e^{-j\omega/2}\cos(\omega/2)$. This is a lowpass (two-point averager) with a zero at $\omega = \pi$. At $\omega = \pi/2$: $H = 0.7071\angle -45^\circ$.

$$y(n) = 0.7071\cos\left(\frac{\pi n}{2} - \frac{\pi}{4}\right)$$

Sinusoid in, the same sinusoid out, scaled by $|H|$ and shifted by $\angle H$.

B19.

Maximum efficiency is where copper loss equals iron loss. Copper loss scales as (load fraction)², so $x^2 \times 45 = 20$ gives $x = \sqrt{20/45} = \mathbf{0.667}$ of full load (two-thirds).

B20.

Assume-and-verify (E2, Type D):

1. Guess a state for each diode (start with the one most obviously forward-biased being ON).
2. Replace each diode with its model: ON is a short, or the V_D source (with r_D for the 3rd model); OFF is an open.
3. Solve the linear circuit.
4. Verify: every ON diode has $i_d > 0$, and every OFF diode has $v_d < V_D$. If any check fails, change that diode's state and repeat.

State the verification explicitly in an exam. That is where the marks are.

B21.

$H(\omega) = \dfrac{1}{1 - 0.9e^{-j\omega}}$. $|H(0)| = 1/0.1 = \mathbf{10}$. $|H(\pi)| = 1/1.9 = \mathbf{0.5263}$. It is **lowpass**. It is stable because the mode 0.9 has $|\lambda| < 1$, so $h(n) = 0.9^n u(n)$ is absolutely summable.

B22.

- **OC:** the test runs at rated voltage, so the flux and the iron loss are at their rated values. The current is only the small magnetising current, so I^2R is negligible.
- **SC:** the test runs at rated current, so the copper loss is at its rated value. The voltage needed is only a few percent of rated, so the flux is tiny. Iron loss scales roughly with $\hat{B}^2 \propto V^2$, so it is negligible.

B23.

- $x(n-3) \leftrightarrow e^{-j3\omega}X(\omega)$. A delay adds linear phase, and the magnitude is unchanged.
- $(-1)^n x(n) = e^{j\pi n}x(n) \leftrightarrow X(\omega - \pi)$. This is modulation (frequency shift) by π, which moves the spectrum around $\omega = 0$ to $\omega = \pi$. A lowpass spectrum becomes **highpass**.

B24.

The diode turns on when v_o exceeds $2 + 0.7 = 2.7$ V. Above that, v_o is held at **2.7 V**. Below 2.7 V, the diode is off and $v_o = v_i$ (no current, so no drop across the resistor). The output is a sine wave with its positive peaks flattened at 2.7 V. The negative half is unchanged, down to −5 V.